

		If resistor is doubled, with same current, voltage across resistor = $0.1 \times (2 \times 10) = 2 \text{ V}$. Hence e.m.f of new battery = $2 + 2 = 4 \text{ V}$
22	A	Option A is false as no current passes through the lower circuit at balance point, hence the internal resistance of cell B does not affect the p.d. across the lower circuit. Option B: When internal resistance in cell A is increased, the potential difference across XY decreases hence balance length XJ increases. Option C: With no current running through cell B, p.d across XJ equals the e.m.f of cell B Option D: As $R \propto l \propto V$, the ratio of length XJ/length XY = pd of XJ/pd of XY = 9/10